



The Federal University of Technology, Akure.
School of Engineering and Engineering Technology
Department of Information & Communication Technology

First Semester Examinations 2023/2024 Session

ICT 501: Java Technology and Programming

Date: May, 2024.

Instructions: Answer all Question Section A and any other three questions from Section B
Course Unit: 2

Time Allowed: 2 ½ Hours

Section A (Compulsory) [40 Marks]

1. What are the values of the following variables? [5 marks]
 - a) double var1 = 10 / 3;
 - b) int var2 = (int) (2.5 * 2.6);
 - c) boolean var3 = !(3 > 3);
 - d) boolean var4 = (121 % 11 == 0) || (5 == 5);
 - e) bint var5 = 11 % 3;
2. Assuming that a and b are both int type variables. Describe the difference between "a = b", "a == b" and "a += b". [3 marks]
3. Suppose you have the variable str of type String with data "How are you?" Write out the data stored in str and place the index of each character below the string. [2 marks]
4. What is the difference between a local variable and an instance variable? [1 Mark]
5. What is the difference between a class and an object? [1 Mark]

6. Study a piece of a java code below and answer all the questions below;

```
int i = 30;
while(i >= 0){
    if(i % 2 > 0){
        continue;
    } System.out.println(i);
    i--;
}
```

- a) What is the output of the above code? [2 Marks]
- b) Rewrite the above code using the for loop [2 Marks]
- c) Rewrite the above code using the do while loop [2 Marks]

7. What is wrong with this piece of code? [3 marks]

```
int count = 0;
while (count < 100)
    if (count % 5 == 0)
        System.out.println(count);
        count++;
```

8. What is wrong with this piece of code? (Circle the error and explain why it's wrong)

[3 marks]

do {

System.out.print("The programming language, "Java",");

```

    }
    while (false);
    System.out.println("is named after the Java coffee");

```

9. What is wrong with this piece of code? (Circle two errors and tell me why they're wrong) [4 marks]

```

public int absoluteValue(int num)
{
    if (num < 0);
        return -num;
    else if (num > 0)
        return num;
}

```

10. Write the output for the piece of code. [8 marks]

```

int a = 2, b = 2, c = 2;
for (int i = 0; i < 3; i++)
{
    a += a;
    b *= b;
    c /= c;
}
System.out.println(a + ", " + b + ", " + c + ".");

```

Handwritten notes:
 a += a; 2 4 8 16
 b *= b; 2 4 16 256
 c /= c; 2 1 1

11. Write the output of the method doSomething(). [8 marks]

```

public void swap(int a, int b)
{
    int temp = a;
    a = b;
    b = temp;
}

```

```

public void doSomething()
{
    int a = 2, b = 3;
    a = b;
    b = a;
    System.out.println(a + ", " + b);
    int c = 2, d = 3;
    int temp = c;
    c = d;
}

```

Handwritten notes:
 a = b; a = 3
 b = a; b = 3
 System.out.println(a + ", " + b); 3, 3
 c = d; c = 3

```

d = temp; d = 2
System.out.println(c + "," + d); 3, 2
int e = 2, f = 3;
swap(f, e);
System.out.println(e + "," + f); 2, 2
}

```

```

System.out.println("input mark:");
mark = Scanner.in

```

import java.util.Scanner

Scanner scanner = new Scanner();

System.out.println("input mark:");

String mark = scanner.next();

Section B [Answer any three [3] question. Each question is worth 20 Marks]

- Write a program in Java that displays the grade of a mark. The grade system is as follows;

90 – 100 A+
80 – 89 A
70 – 79 B+
60 – 69 B
50 – 59 C+
45 – 49 C
0 – 44 D

90 ≥ mark ≥ 100

Your program must not allow marks below 0 and above 100. Use the command line both for prompting the user for a mark and displaying the grade [20 Marks]

- Create a class that holds 10 grades in an array data structures. [2 Marks]

- Loop through the data structure and compute the sum of grades [2 Marks]
- Loop through the data structure and compute the average of grades [4 Marks]
- Loop through the data structure and print the biggest grade [4 Marks]
- Loop through the data structure and print the lowest grade [4 Marks]
- Print the grades in the descending order of indexes [4 Marks]

for (i = 0; i < 10; i++)

- Write an instantiable class called Counter that has:

- An integer instance variable called numClicks [2 Marks]
- A constructor that sets numClicks to 0 [2 Marks]
- A method called click() that adds 1 to numClicks [4 Marks]
- A method called getClicks() that returns numClicks [4 Marks]

sum = 0

numClicks = 0

click()

Write a driver program called TestCounter that

- Creates a counter object [2 Marks]
- Calls the click method 2 times [4 Marks]
- Calls the getClicks() method and prints its value [2 Marks]

sum = grades[i]

Counter

Ans: numClicks

Counter counter = new Counter();

numClicks = 0

- Write a main method that

- prompts a user to enter an integer [2 Marks]
- passes that integer to a method called cube [2 Marks]
- receives the cube of that integer back from the method [2 Marks]

Counter c

?

216 120

12121
m
121

d) displays the return value

[2 Marks]

Write a method called cube that

e) accepts an integer, cubes it and returns the result

[4 Marks]

f) Change the preceding code so the main method has a *for* loop

[4 Marks]

g) It passes the numbers 0 to 3 to the cube method, one at a time

[2 Marks]

h) It receives back the result from the cube method and displays it

[2 Marks]

i) The cube method is unchanged

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THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
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DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING

B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT501 **COURSE TITLE:** Java Technology and Programming

UNITS: 2

TIME ALLOWED: 2½ Hours

Instruction: Read Carefully and Answer any Three(3)

- 1a. ✓ Java is a full-fledged and powerful language that can be used in many ways. It comes in three editions. List and explain the three editions. Also, provide the difference between API, JDK, and IDE [10Marks]
- 1b. ✓ What is Object-Oriented Programming (OOP). Define polymorphism in Java and differentiate between compile-time polymorphism and run-time polymorphism. Explain the concepts of encapsulation and abstraction in Java. [10Marks]
- 2a. ✓ Using the short code below, in tabular format, provide the following: the class name, the method, the output of the code, the special characters, the research words, extract the method block, the class block, the line comment and the statement terminator. [10Marks]

```
1 public class WelcomeClass300 {  
2     public static void main(String[] args) {  
3         // Display message Welcome to Java! on the console  
4         System.out.println("Welcome to CSC313!");  
5     }  
6 }
```

- 2b. ✓ A program can decide which statements to execute based on a condition. Differentiate the following selection methods *if statement*, *two-way if-else statement* and *Nested if statement*. Provide example for each. [10Marks]

- 3a. ✓ Using UML notation, design a class diagram for a TemperatureSensor class that could be used in a circuit board temperature monitoring system. The class should include the following:
- o Attributes: sensorID (String), currentTemperature (double), maxSafeTemperature (double)
 - o Methods: getCurrentTemperature() (double), checkTemperatureSafety() (boolean), setMaxSafeTemperature(double maxTemp) (void)
 - o Constructor: TemperatureSensor(String sensorID, double maxSafeTemperature)

Draw the UML class diagram and explain how each component (class name, attributes, methods, and constructor) is represented in the diagram. Also explain how this class could be used in a circuit board temperature monitoring system. [8Marks]

- 3b. ✓ Explain the concept of inheritance in Java with an example related to power distribution grid analysis. Create a superclass PowerComponent with a method checkStatus() and a subclass Transformer that inherits from PowerComponent and adds a method monitorLoad(). Write the Java code for both classes and demonstrate how the Transformer class can use the checkStatus() method from the PowerComponent class. [12Marks]

- 4a. i. ✓ Declare on one line and instantiate on a second line a one-dimensional array called strArr that holds 50 Strings. [2Marks]
- ii. ✓ Declare on one line and instantiate on a second line a two-dimensional array called intArr that holds int values and has 5 rows and 8 columns. [4Marks]



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- 4b. A smart energy monitoring system tracks energy consumption in 3 industrial zones across 7 days. The data is stored in a 2D array where rows represent zones and columns represent days.

Write a Java program to:

1. Calculate the weekly energy consumption for each zone.
2. Identify the day with the highest total energy consumption.

Sample Data (Energy Consumption in kWh):

`int[][] energyData = {{500, 520, 540, 510, 495, 530, 515}, {610, 625, 600, 590, 580, 605, 620}, {720, 710, 735, 725, 705, 740, 715}};` [14Marks]

- 5a. What will be the output of the following codes [9Marks]

a. `double[] dblArr = {3.5, 6.8, 2.3, 9.1, 1.0};`
`for (int i = 0; i < dblArr.length; i++) { dblArr[i] /= 2; }`
`for (int i = dblArr.length - 1; i >= 0; i--) {`
`System.out.println("Value: " + dblArr[i]);`

b. `int[] intArr = {1, 2, 3, 4, 5};`
`double[] dblArr = {.5, 1, 1.5, 2, 2.5};`
`for (int i = 0; i < intArr.length; i++) {`
`dblArr[i] = dblArr[i] * intArr[i];`
`}`
`for (int i = 0; i < intArr.length; i++) {`
`System.out.println(intArr[i] + " : " + dblArr[i]);`

c. `int[] intArr = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};`
`for (int i = 0; i < intArr.length; i += 3) {`
`System.out.println("Value: " + intArr[i]);`
`}`

- 5b. An EV charging station records the charging time (in minutes) of vehicles at five charging ports across four time slots in a day. The data is stored in a 2D array where rows represent ports, and columns represent time slots. Write a Java program to:

1. Compute the total charging time for each port.
2. Find the charging port with the highest usage.

Sample Data (Charging Time in Minutes):

`int[][] chargingTime = {{120, 135, 110, 140}, {150, 130, 145, 160}, {100, 120, 130, 115}, {140, 155, 135, 145}, {135, 140, 150, 155}};` [11Marks]

```
int[][] chargingTime {
    {120, 135, 110, 140}
    {150, 130, 145, 160}
    {100, 120, 130, 115}
    {140, 155, 135, 145}
    {135, 140, 150, 155}
}
```

```
int total = 0; HighestUsage = 0; Port = 0;
for (int i = 0; i < chargingTime.length; i++) {
    for (int k = 0; k < chargingTime[i].length; k++) {
        int value = chargingTime[i][k];
        total += value;
    }
    print total for i;
    if (total > HighestUsage) {
        HighestUsage = total;
        Port = i;
    }
}
```

Print